

Technical Document #644: Data Engineering - Comprehensive Overview

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Partitioning divides data into smaller, more manageable pieces. It improves query performance by reducing the amount of data scanned.

Data governance establishes policies and procedures for managing data assets. It covers data privacy, security, and compliance.

Data pipelines automate the movement and transformation of data between systems. Apache Airflow and Prefect are popular workflow orchestration tools.

Augmented reality overlays digital information on the physical world. It has applications in training, maintenance, and entertainment.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Key Takeaways:

- Stream processing handles continuous data streams in real-time.
- Schema evolution handles changes in data structure over time.
- Data lakes store raw data in its native format.